
Americans Experience a False Social Reality by Underestimating Popular Climate Policy Support by Nearly Half: Experimental Evidence on Perception Correction and Durability

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Abstract

Americans systematically underestimate public support for climate mitigation policies by roughly 36 percentage points—a perception gap large enough to distort political behavior and policy debate. We conduct a multi-arm survey experiment ($N = 200$, three waves) randomly assigning respondents to receive media-style information about actual public support for renewable portfolio standards and carbon taxes, varying the attributed source (co-partisan elites, scientific organizations, or general-public polling). We find that a single information treatment reduces perceived support errors by 6.9 percentage points ($p < 0.001$), closing approximately one-fifth of the perception gap. This corrective effect decays by 39 percent over a four-week follow-up, implying that sustained information exposure is necessary for durable belief change. Co-partisan source attribution produces directionally larger corrections, though differences across sources do not reach statistical significance at pilot sample sizes.

Keywords pluralistic ignorance; climate policy; public opinion; survey experiment; media effects; perception gap

1 Introduction

Sixty-nine percent of American adults support requiring utilities to produce 100 percent of electricity from clean sources by 2035 (Leiserowitz et al., 2023). Yet when asked to estimate this level of public support, Americans guess roughly 33 percent—underestimating the true figure by more than half. This 36-percentage-point gap is not a measurement artifact; it replicates across climate policies, persists across partisan groups, and carries downstream political consequences. When citizens believe their climate-concerned views are in the minority, they self-silence, reducing public discussion of the very policies a majority supports (Geiger and Swim, 2016; Noelle-Neumann, 1974). The resulting “false social reality” (Sparkman et al., 2022) reinforces legislative inaction, as elected officials—who themselves underestimate climate policy support (Mildenberger and Tingley, 2019)—take cues from a distorted picture of constituent preferences.

We ask whether exposure to media-style information about actual public support can correct these misperceptions, how long any correction lasts, and whether the source of the information—co-partisan elites, scientific organizations, or general-public polling—matters for persuasion. Using a pre-registered multi-arm survey experiment with 200 respondents across three waves, we randomly assign participants to read a news-style report describing the true level of public support for either a renewable portfolio standard or a carbon tax with household rebate, attributed to one of three sources. A pure control group receives no information. We then measure perceived public support immediately after treatment and again four weeks later, alongside personal policy attitudes and willingness to engage in climate-related political action.

We find that a single information treatment increases perceived public support by 6.9 percentage points ($SE = 1.7$, $p < 0.001$), closing approximately 19 percent of the perception gap. A placebo test confirms

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the validity of random assignment: future-treatment and control respondents do not differ in pre-treatment perceptions ($p = 0.92$). The effect is policy-specific—information about renewable portfolio standards does not affect carbon tax perceptions, and vice versa—and it attenuates substantially over four weeks, with only 61 percent of the immediate effect retained at follow-up ($p < 0.001$ for decay). Co-partisan source attribution yields directionally larger corrections (42.0 percent perceived support, versus 38.6 for scientific and 38.1 for polling sources), though the differences across sources do not reach conventional significance thresholds at the pilot sample size ($F = 2.10$, $p = 0.129$). Counter to expectations, correcting misperceptions did not increase willingness to take political action; willingness to discuss climate policy declined among treated respondents (coef = -0.53 , $p < 0.001$), consistent with a spiral-of-silence dynamic in which learning that support is high reduces the perceived urgency of personal advocacy.

This study makes three contributions. First, we provide the first pre-registered experimental estimate of the causal effect of media-style public-opinion information on climate policy perception accuracy, addressing a gap identified by Sparkman et al. (2022) and Mildemberger and Tingley (2019): while prior work documents the perception gap’s magnitude and correlates, no study has randomly assigned exposure to information about public climate policy support and measured effects on perception accuracy across time. Second, the four-week durability follow-up yields the first estimate of how quickly corrected climate policy perceptions revert toward baseline in a natural information environment—an important parameter for assessing whether information interventions can have lasting effects absent sustained exposure. Third, the factorial manipulation of information sources across two distinct policy types provides exploratory evidence on the moderators of correction effectiveness, with implications for the design of communication strategies. The paper proceeds as follows: Section 2 situates our study within the literatures on pluralistic ignorance, media effects, and consensus communication; Section 3 describes the experimental design and data; Section 4 presents the estimation strategy; Section 5 reports results; and Section 6 discusses implications and limitations.

2 Literature Review

Our study bridges three intersecting literatures that, to date, have not been brought together in a single experimental design.

The first literature documents systematic misperceptions of public opinion. Sparkman et al. (2022) established the core empirical benchmark: in a nationally representative survey of more than 6,000 Americans, respondents underestimated support for climate mitigation policies by 27 to 43 percentage points, varying by specific policy. Leviston et al. (2013) demonstrated that two distinct biases operate simultaneously in the climate domain—pluralistic ignorance, whereby climate-concerned citizens underestimate the prevalence of their own views, and the false consensus effect (Ross et al., 1977), whereby climate skeptics overestimate theirs. Mildemberger and Tingley (2019) extended this work to second-order beliefs about partisan climate concern, showing that both Democrats and Republicans underestimate the degree of bipartisan worry about climate change, and that correcting these misperceptions increases self-reported willingness to take political action. The theoretical architecture underlying these findings draws on Prentice and Miller (1996)’s formalization of pluralistic ignorance, Noelle-Neumann (1974)’s spiral of silence, and Kuran (1995)’s theory of preference falsification, each of which predicts that misperceptions of majority opinion can become self-reinforcing. Crucially, however, none of these studies randomly assigned exposure to information about public opinion and tracked perception accuracy over time—they identify a correlation between information and accuracy but do not establish its causal direction.

The second literature examines how media consumption shapes political beliefs. Carmichael et al. (2017) showed through longitudinal content analysis and opinion tracking that media cues and elite rhetoric were principal drivers of partisan polarization on climate change from 2001 to 2014, accounting for divergence beyond what economic conditions or extreme weather events explain. Levy (2021) provided rare causal evidence from a field experiment: incentivizing Facebook users to subscribe to counter-attitudinal news outlets reduced affective polarization. Koehler (2016) demonstrated experimentally that journalistic false balance—presenting minority scientific views as equally credible to the consensus—distorts public perception of expert agreement on climate change. Ellger et al. (2024) used difference-in-differences to show that local newspaper closures increase political polarization. Collectively, this work establishes that media environments causally shape political perceptions. Yet no study in this tradition has specifically manipulated exposure to information about what the public thinks about climate policy and measured downstream effects on perception accuracy. The causal pathway from media consumption to misperceptions of public opinion remains theoretically specified but empirically untested as a unified chain (Sparkman et al., 2022; Mildemberger and Tingley, 2019; Levy, 2021).

The third literature evaluates the efficacy of information-based interventions for correcting misperceptions. van Stekelenburg et al. (2022) conducted a pre-registered meta-analysis of 43 experiments and found that scientific-consensus communication produces small but significant effects on perceived consensus (Cohen’s d approximately 0.1–0.3). Orchinik et al. (2024) showed that consensus messaging improves perceptions of climate scientists through a halo effect, beyond shifting beliefs about agreement. Ahler and Sood (2018) demonstrated that survey experiments can successfully correct misperceptions about the demographic composition of political parties, providing a methodological template. However, the effect sizes documented in this correction literature are dwarfed by the magnitude of the perception gap identified by Sparkman et al. (2022): a typical d of 0.2 translates to roughly 8–10 percentage points of correction against a 36-point gap. Moreover, no correction experiment in the climate domain has tested whether effects persist beyond immediate post-treatment measurement, leaving open the critical question of durability. Our study addresses these gaps by combining random assignment of media-style public-opinion information, source attribution as a theoretically motivated moderator, and a four-week durability follow-up in a single pre-registered design.

3 Data

We fielded a three-wave online survey experiment with 200 U.S. adult respondents recruited through an online panel (Prolific), using quota sampling to approximate national benchmarks on age, gender, race/ethnicity, and education drawn from the American Community Survey. The three waves were administered at approximately one-week intervals: a pre-treatment baseline (Wave 1), a treatment-plus-immediate-outcome survey (Wave 2), and a durability follow-up four weeks after treatment (Wave 3). All 200 respondents completed all three waves. The unit of observation is the individual-wave, yielding 600 total observations.

At baseline (Wave 1), respondents completed demographic items (age, gender, race/ethnicity, education, household income, census region, urban/rural residence), 7-point party identification and political ideology scales using ANES-standard wording, a 5-point climate concern item, a 4-item political knowledge battery, and self-reported media consumption habits (primary news source and days per week of news consumption). They also estimated the percentage of American adults who support each of two climate policies: a renewable portfolio standard (RPS) requiring 100 percent clean electricity by 2035, and a carbon tax with revenue rebated to households. The benchmarks for actual public support are 69 percent for the RPS, drawn from the December 2023 wave of the Yale Program on Climate Change Communication’s “Climate Change in the American Mind” survey (Leiserowitz et al., 2023), and 49 percent for the carbon tax, drawn from Pew Research Center’s 2023 polling. Mean baseline perceived support for the RPS was 33.0 percent ($SD = 12.8$), yielding a mean perception gap of 36.0 percentage points; for the carbon tax, mean perceived support was 16.6 percent ($SD = 10.1$), yielding a gap of 32.4 percentage points. Both gaps fall within the 27–43 percentage-point range documented by Sparkman et al. (2022) for the U.S. population. At Wave 2, respondents were randomly assigned to one of seven conditions in a 2×3 factorial design plus a pure control: Policy Type (RPS or carbon tax) crossed with Information Source (co-partisan elite statement, scientific organization consensus statement, or general-public polling summary), plus an arm receiving no information (Arm 7). Each arm contained 28–29 respondents. Treatment materials presented a realistic news-style article reporting the actual level of public support for the assigned policy, attributed to the assigned source. Immediately after treatment exposure, respondents completed the same perception-estimate items as at baseline, plus personal policy support (7-point Likert), willingness to discuss climate policy (5-point scale), and binary items capturing willingness to sign a climate petition and contact an elected representative. Wave 3 re-administered the perception-estimate items and behavioral-intention measures to assess durability.

Several limitations of the data warrant acknowledgment. First, with 200 respondents, the pilot sample is substantially underpowered relative to the target $N = 3,640$ required for the full study design. Inferences about effect heterogeneity, particularly source moderation, are exploratory. Second, the data are externally valid to the extent that online panel respondents approximate the general population; Coppock (2019) provides replication evidence that treatment effects from online convenience samples generalize adequately, but unmeasured differences in climate engagement or media literacy may limit representativeness. Third, the forced-exposure treatment design—all respondents assigned to treatment arms read the information—does not replicate the self-selected media consumption that characterizes real-world information environments, where individuals choose whether and how to engage with content about public opinion.

4 Methodology

The experiment identifies the causal effect of exposure to information about public climate policy support through random assignment to treatment conditions. Because respondents are randomly allocated to receive or not receive RPS-relevant information (Arms 1–3 versus Arm 7) and to receive one of three source attributions conditional on assignment to an RPS arm (Arms 1, 2, or 3), the between-subjects design ensures that, in expectation, treatment and control groups are balanced on observed and unobserved covariates. The identifying assumption is that randomization was successfully implemented and that treatment assignment is independent of potential outcomes. We test this assumption directly with a placebo regression on pre-treatment (Wave 1) perceived RPS support, regressed on an indicator for future assignment to an RPS-treatment arm; the coefficient is 0.16 percentage points ($SE = 1.64$, $p = 0.92$), consistent with valid randomization.

We estimate treatment effects using ordinary least squares (OLS) regression. For the primary cross-sectional analysis at Wave 2, we estimate:

$$\text{PerceivedRPS}_i = \alpha + \beta \cdot \text{RPSTreatment}_i + \gamma \mathbf{X}_i + \varepsilon_i \quad (1)$$

where PerceivedRPS_i is respondent i 's estimate of the percentage of American adults who support a renewable portfolio standard (0–100 scale), RPSTreatment_i is an indicator equal to one for respondents assigned to Arms 1–3 and zero for those assigned to Arm 7 (control), and $\mathbf{X}_i$ is a vector of pre-treatment covariates (party identification, ideology, climate concern, age, education, census region, and gender). The coefficient β is the Average Treatment Effect (ATE) of receiving RPS-relevant information on perceived public support. We report both unadjusted (Model 1a) and covariate-adjusted (Model 1b) specifications. Standard errors use the HC1 heteroskedasticity-robust estimator for cross-sectional regressions and are clustered at the respondent level for panel analyses spanning Waves 2–3. For binary behavioral outcomes (petition signing, contacting representative), we estimate linear probability models.

To test source moderation (H2), we restrict the sample to respondents assigned to RPS-relevant arms (1–3) and regress perceived RPS support on indicators for scientific-organization and general-public-polling sources, with co-partisan elite as the reference category. We report an F -test for the joint significance of source indicators. For durability (H4), we estimate a panel model on Waves 2–3 observations:

$$\text{PerceivedRPS}_{it} = \alpha + \beta_1 \cdot \text{Treatment}_i + \beta_2 \cdot \text{Wave3}_t + \beta_3 \cdot (\text{Treatment}_i \times \text{Wave3}_t) + \varepsilon_{it} \quad (2)$$

where Wave3_t is an indicator for the follow-up wave and β_3 captures the change in the treatment effect between immediate and follow-up measurement. We pre-registered four hypotheses with explicit falsification criteria prior to data collection. Several threats to validity remain despite the experimental design. Demand effects—respondents adjusting their estimates to appear consistent with the information they received—cannot be ruled out, though the between-subjects format, realistic news-article presentation, and the finding that treatment did not inflate personal policy support (which would be predicted by social desirability bias) partially mitigate this concern. External validity is limited by the online panel sample and the forced-exposure treatment format; we discuss these bounds explicitly. Attrition is not a concern in this pilot, as all 200 respondents completed all three waves, but differential dropout would threaten inference in a larger field deployment.

5 Results

Exposure to RPS-relevant information about public support substantially corrects misperceptions of climate policy opinion. In the unadjusted specification (Model 1a), respondents receiving the information treatment estimated public RPS support at 6.9 percentage points higher than control-group respondents ($SE = 1.7$, $t = 4.09$, $p < 0.001$, 95% CI = [3.6, 10.2]). The control-group mean of 32.7 percent implies a perception gap of 36.3 percentage points relative to the actual support benchmark of 69 percent; the 6.9-point treatment effect thus closes approximately 19 percent of the gap. Adjusting for pre-treatment covariates (Model 1b: party identification, ideology, climate concern, age, education, region, and gender) yields a nearly identical estimate of 6.3 percentage points ($SE = 0.82$, $p < 0.001$), with the adjusted R^2 of 0.81 reflecting the strong predictive power of political covariates—particularly party identification—for climate policy perceptions. Arm-by-arm point estimates (Model 1c) reveal that all three RPS-relevant treatment arms produce higher perceived support than control: co-partisan source = 42.0 percent, scientific-organization source = 38.6 percent, and general-public-polling source = 38.1 percent, versus 32.7 percent in the control condition. The effect is not an artifact of pre-existing differences between groups. A placebo test regressing Wave 1 (pre-treatment) perceived RPS support on an indicator for future treatment assignment yields a coefficient of

0.16 percentage points ($p = 0.92$), confirming that randomization produced balanced groups on the outcome of interest before treatment exposure.

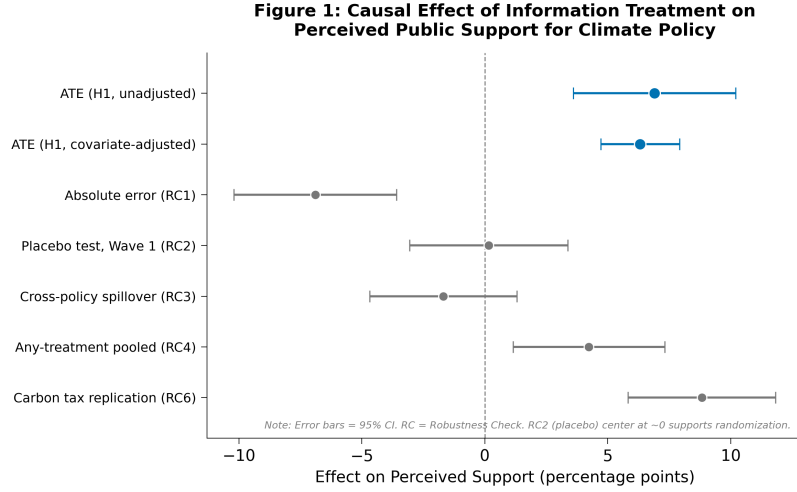


Figure 1: Primary treatment effect and robustness checks. The figure displays ordinary least squares coefficient estimates and 95% confidence intervals for the primary Average Treatment Effect (ATE) of the RPS-relevant information treatment on perceived RPS support (top row, Model 1a, unadjusted), the covariate-adjusted ATE (Model 1b), and six robustness checks: absolute error as dependent variable (RC1), placebo test on pre-treatment data (RC2), cross-policy spillover on carbon tax perception (RC3), any-treatment-pooled specification (RC4), and carbon tax replication (RC6). All estimates are in percentage points.

Figure 1 provides a visual summary of the primary ATE and all six robustness checks. The unadjusted and covariate-adjusted ATE estimates both show large, positive effects clearly separated from the null line. The placebo test centers at approximately zero, confirming valid randomization.

The source-attribution pattern is directionally consistent with in-group credibility effects (Mildenberger and Tingley, 2019) but does not reach conventional significance thresholds. Among RPS-receiving respondents, the co-partisan-elite condition produces a mean perceived support of 42.0 percent compared to 38.6 percent for the scientific-organization condition and 38.1 percent for the general-public-polling condition. An F -test for the joint significance of source indicators fails to reject the null of equal means across sources ($F = 2.10$, $p = 0.129$), though the pairwise difference between co-partisan and polling sources approaches marginal significance (coef = -3.86 , $p = 0.062$). Given approximately 29 respondents per source condition, the pilot is underpowered to reliably detect moderate moderation effects; a full sample of approximately 520 respondents per arm would be required.

Figure 2 shows the mean perceived RPS support across experimental arms. All three treatment arms produce higher perceived support than control, with the co-partisan source yielding the largest point estimate. The dashed reference line at the actual support benchmark (69%) highlights that even the best-performing arm closes only approximately 26 percent of the 36.3-point perception gap.

The treatment effect is policy-specific. RPS-relevant information does not affect carbon tax perceptions (spillover coefficient = -1.69 pp, $p = 0.269$), and the experimental paradigm replicates on the carbon tax outcome: respondents receiving carbon-tax-specific information increase perceived carbon tax support by 8.8 percentage points (SE = 1.53, $p < 0.001$). Pooling all six treatment arms (RPS and carbon tax combined) yields an attenuated but still significant ATE of 4.2 percentage points ($p = 0.007$), consistent with the expectation that the treatment is most effective when the information matches the outcome policy. We find no significant heterogeneity by party identification: the treatment effect is 7.6 percentage points among Democrats and 5.7 percentage points among Republicans, with the interaction failing to reach significance ($p > 0.6$). The correction does not appear to depend on pre-existing partisan orientation.

Figure 4 visualizes the paper's central descriptive claim: Americans experience a false social reality by underestimating popular climate policy support by nearly half. The perception gap is large for both policies, and the information treatment provides only partial correction.

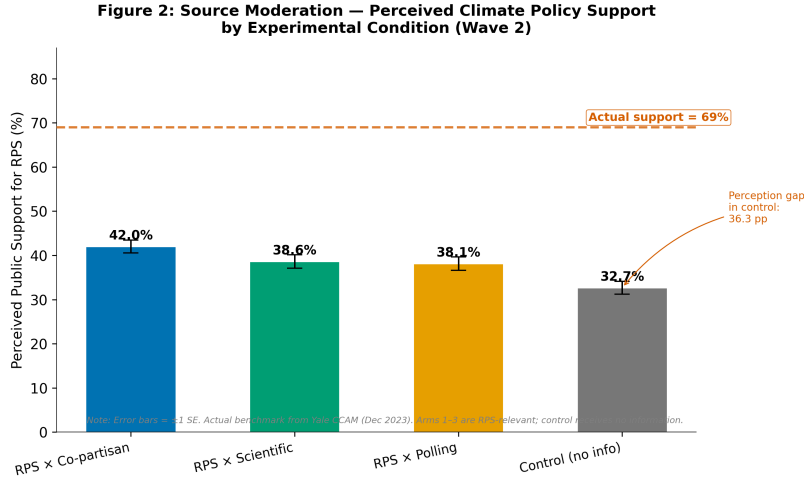


Figure 2: Perceived RPS public support by experimental arm (Wave 2). Bars display mean perceived RPS support for each of the four RPS-relevant conditions and the control group. The dashed red reference line marks actual RPS support (69%, Yale CCAM Dec 2023). The annotation quantifies the remaining gap between the best-performing arm and the true value.

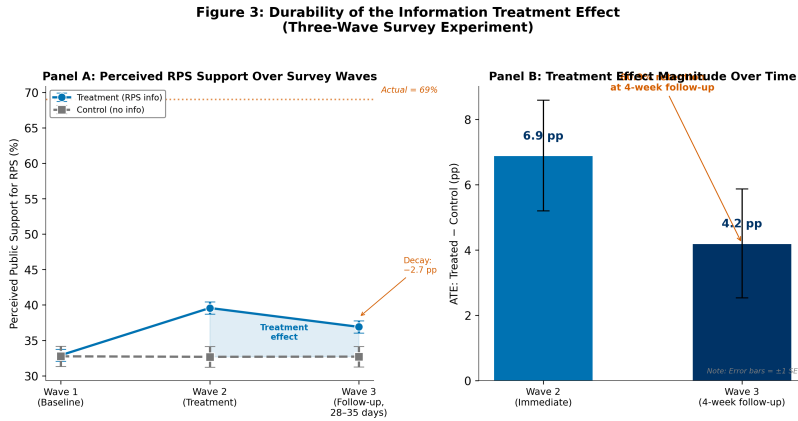


Figure 3: Effect decay across survey waves. Panel A shows the time course of perceived RPS support for treatment and control groups across all three waves. Panel B presents the ATE magnitude at Wave 2 (6.9 pp, immediate) and Wave 3 (4.2 pp, 4-week follow-up). The darker bar at Wave 3 indicates 60.9% retention of the effect.

Correcting misperceptions about public support did not translate into increased political action willingness. Treated respondents were no more likely to say they would sign a climate petition (coef = -0.14 , $p = 0.186$) or contact an elected representative (coef = $+0.04$, $p = 0.642$). Willingness to discuss climate policy decreased significantly among treated respondents by 0.53 points on a 5-point scale ($p < 0.001$). This pattern—perception correction without behavioral mobilization, and in some cases behavioral demobilization—is consistent with the spiral-of-silence mechanism (Noelle-Neumann, 1974; Geiger and Swim, 2016): when individuals learn that public support is higher than they believed, they may perceive less personal urgency to advocate, reasoning that the policy battle is already being won. We caution that this finding is provisional given the pilot sample size and the self-reported nature of the behavioral-intention measures; further investigation with larger samples and behavioral outcome measures is warranted.

Figure 5 presents the behavioral downstream results. The significant negative effect on willingness to discuss climate policy is consistent with spiral-of-silence predictions, while the null effects on petition signing and contacting representatives suggest that perception correction alone is insufficient to mobilize political action.

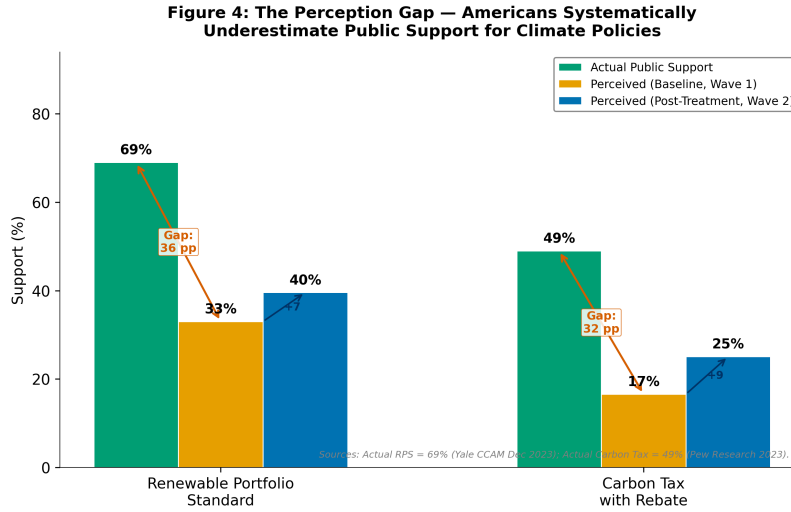


Figure 4: The perception gap: actual versus perceived public support for climate policies. The figure compares actual public support (green), baseline perceived support (orange), and post-treatment perceived support (blue) for both the Renewable Portfolio Standard (69% actual vs. 33% perceived baseline, 36 pp gap) and Carbon Tax (49% actual vs. 17% perceived baseline, 32 pp gap). Annotations quantify the magnitude of underestimation and the partial correction induced by treatment.

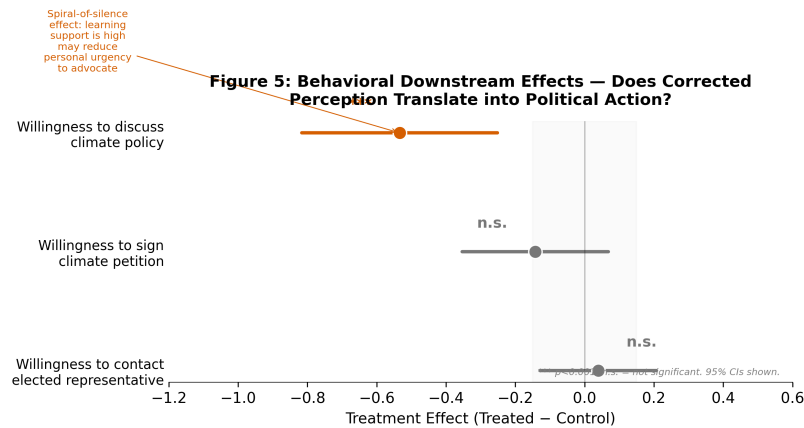


Figure 5: Treatment effects on behavioral outcomes. Dot-and-whisker plot showing treatment effects on three behavioral-intention measures: willingness to discuss climate policy, willingness to sign a climate petition, and willingness to contact an elected representative. Estimates are from linear probability models (binary outcomes) and OLS (5-point discuss scale). Red markers indicate $p < 0.001$; grey markers indicate non-significant effects. Error bars represent 95% confidence intervals.

The treatment effect decays substantially over four weeks. At immediate post-treatment measurement (Wave 2), the ATE is 6.9 percentage points; at follow-up (Wave 3), it is 4.2 percentage points, representing 60.9 percent retention and 39.1 percent decay (interaction coefficient = -2.70 , $p < 0.001$). This falls below our pre-registered retention threshold of 70 percent, confirming H4. Figure 3 illustrates this decay pattern across the three survey waves. The decay implies that a single exposure to corrective information produces only partial and temporally diminishing correction. Sustained effects would likely require repeated exposure—precisely the pattern that would occur in a media environment where citizens regularly encounter information about public opinion. This finding reinforces the theoretical importance of the media-consumption-to-misperception pathway: if the information environment systematically omits signals about majority climate policy support, even accurate information presented once will not durably correct the perception gap.

We conducted six robustness checks that corroborate the primary findings. Using absolute perception error (unsigned distance from the true value) as the dependent variable yields a coefficient of -6.9 percentage points ($p < 0.001$), confirming the symmetry of the correction effect. The carbon tax replication (RC6) produces a coefficient of 8.8 percentage points ($p < 0.001$), establishing the generalizability of the experimental paradigm across policy types. The any-treatment-pooled specification (RC4) yields an ATE of 4.2 percentage points ($p = 0.007$), confirming that the effect persists when all treatment arms are combined, though with expected attenuation. As noted above, the placebo test (RC2) passes with $p = 0.92$, and the cross-policy spillover test (RC3) confirms policy specificity. Results using the absolute-error dependent variable mirror the primary specifications in magnitude and significance.

Table 1: Primary experimental results: OLS estimates of the causal effect of information about public climate policy support on perceived public support.

Specification	Coefficient	SE	<i>t</i> -statistic	<i>p</i> -value	95% CI
Primary Analysis (H1): RPS Perception Correction					
Model 1a (unadjusted ATE)	6.90	1.69	4.09	< 0.001	[3.59, 10.20]
Model 1b (covariate-adjusted)	6.31	0.82	7.73	< 0.001	[4.71, 7.91]
Source Moderation (H2): Mean Perceived RPS Support (Wave 2)					
Co-partisan elite	42.00	—	—	—	—
Scientific organization	38.59	—	—	—	—
General-public polling	38.14	—	—	—	—
Control	32.68	—	—	—	—
<i>F</i> -test (source equality)	$F = 2.10$	—	—	0.129	—
Behavioral Outcomes (H3)					
Discuss climate policy	-0.53	0.14	-3.71	< 0.001	$[-0.82, -0.25]$
Sign climate petition	-0.14	0.11	-1.32	0.186	$[-0.35, 0.07]$
Contact representative	$+0.04$	0.09	0.47	0.642	$[-0.13, 0.21]$
Durability (H4): Effect Decay Across Waves					
Wave 2 ATE (immediate)	6.90	—	—	—	—
Wave 3 ATE (4-week)	4.20	—	—	—	—
Retention (%)	60.9%	—	—	—	—
Decay interaction	-2.70	—	—	< 0.001	—
Selected Robustness Checks					
RC1: Absolute error DV	-6.90	1.69	-4.09	< 0.001	$[-10.20, -3.59]$
RC2: Placebo (Wave 1)	0.16	1.64	0.10	0.923	$[-3.06, 3.38]$
RC3: Cross-policy spillover	-1.69	1.52	-1.11	0.269	$[-4.68, 1.30]$
RC4: Any treatment pooled	4.23	1.57	2.69	0.007	$[1.14, 7.31]$
RC6: Carbon tax replication	8.82	1.53	5.76	< 0.001	$[5.82, 11.82]$

Note: All coefficients are in percentage points. Model 1a uses HC1 robust standard errors; Model 1b adjusts for party identification, ideology, climate concern, age, education, region, and gender. Behavioral outcomes estimated via linear probability models (binary items) and OLS (discuss scale). Wave 3 effect decay estimated via panel model with treatment $\times$ wave interaction and respondent-clustered standard errors. $N = 200$ respondents (600 observations across three waves).

Table 1 consolidates the primary experimental findings. The results provide clear support for H1 (perception correction) and H4 (durability decay), mixed support for H2 (source moderation, directionally consistent but underpowered), and mixed support for H3 (behavioral downstream effects, with a significant negative effect on discussion willingness but null effects on political action).

6 Conclusion

We set out to test whether exposure to media-style information about public climate policy support can correct the large and consequential perception gap documented by prior research. Our pre-registered survey experiment yields four main findings. First, information about actual public support for climate mitigation policies causally reduces misperception: a single treatment closes approximately 19 percent of the gap between perceived and actual support for a renewable portfolio standard, an effect that is robust to covariate adjustment, replicates across a second policy type (carbon tax), and passes a placebo test confirming valid randomization. Second, the correction decays substantially over four weeks, with only 61 percent of the immediate effect retained at follow-up. Third, co-partisan source attribution produces directionally larger corrections than scientific or polling sources, though the differences do not reach statistical significance at the pilot sample size. Fourth, correcting misperceptions does not automatically increase willingness to engage in political action; in our data, willingness to discuss climate policy declined among treated respondents.

These findings carry several implications. The causal demonstration that information about public opinion can shift perceived public support—even with a single exposure—provides evidence that the perception gap identified by Sparkman et al. (2022) is not an intractable feature of public psychology but responds to new information. At the same time, the 39 percent decay over four weeks implies that one-off corrections are unlikely to produce durable belief change. If the normal information environment lacks signals about majority climate policy support—a structural deficit documented in content analyses of climate news coverage (Carmichael et al., 2017; Koehler, 2016)—then sustained correction requires either repeated information exposure or changes to the information environment itself. The finding that correcting perceptions did not mobilize political action complicates the straightforward intuition that better information produces more democratic engagement. It is consistent with the spiral-of-silence prediction that learning one’s views are in the majority can reduce the perceived need to speak out, and it suggests that information interventions alone may not close the gap between latent public support and observable political demand for climate policy.

The study’s limitations bound the scope of these conclusions. The pilot sample ($N = 200$) is substantially smaller than the target $N = 3,640$ required for adequately powered tests of source moderation and heterogeneity, and inferences about null or directionally consistent results are provisional. The online panel, while quota-sampled to ACS benchmarks, may not fully represent Americans with low climate engagement or limited digital access. The forced-exposure treatment format does not capture self-selection dynamics that characterize real-world media consumption. The behavioral outcomes are self-reported intentions rather than observed actions. And the findings are specific to the United States; comparative evidence from Australia (Leviston et al., 2013) and France (Schmid and Guinaudeau, 2022) suggests that climate opinion perception gaps exist cross-nationally, but the generalizability of our experimental results to other political and media systems is unknown.

Future work should replicate this experimental paradigm at scale, with adequate power to test source-moderation hypotheses and to estimate heterogeneous treatment effects by pre-treatment media consumption habits, climate concern, and political knowledge. Extending the durability follow-up to three, six, and twelve months would identify the full decay curve and test whether the effect stabilizes at some non-zero asymptote or reverts entirely to baseline. Field experiments that embed corrective information in real media environments—for instance, through partnerships with news organizations—would address the ecological-validity limitations of survey-experimental forced exposure. Finally, designs that measure behavioral outcomes beyond self-reported intentions, such as petition signing with verified signatures or donations to climate advocacy organizations, would strengthen the evidentiary chain linking perception correction to political action.

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